

Research/Biology Role – Physiology Essentials

1. Cardiovascular System (Heart & Blood Flow)

- **Heart Anatomy:** Four chambers (two atria, two ventricles) pump blood in a loop — right side pumps deoxygenated blood to the lungs, left side pumps oxygenated blood to the body.
 - **Heart Rate (HR):** Beats per minute, controlled by the sinoatrial (SA) node (pacemaker). Normal adult range: **60–100 bpm**.
 - **Blood Pressure (BP):**
 - **Systolic** (higher, when ventricles contract).
 - **Diastolic** (lower, when ventricles relax).
 - Typical range: ~120/80 mmHg.
 - **Relevance for Devices:**
 - **ECG/EKG** → monitors electrical activity of the heart.
 - **Blood pressure cuffs** → measure pressure using occlusion + sensors.
 - **Wearables** → track HR through PPG or ECG electrodes.
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2. Respiratory System (Breathing & Gas Exchange)

- **Breathing Mechanics:** Diaphragm contracts → lungs expand → air in; diaphragm relaxes → lungs compress → air out.
- **Tidal Volume:** Air exchanged in normal breathing (~500 mL).

- **Respiratory Rate (RR):** Breaths per minute, normal adult range: **12–20**.
 - **Gas Exchange:**
 - Oxygen diffuses from alveoli → blood.
 - CO₂ diffuses from blood → alveoli.
 - **Relevance for Devices:**
 - **Spirometers** → measure airflow and lung capacity.
 - **Sleep apnea devices** → track irregular breathing/snoring.
 - **Ventilators** → mechanically assist breathing.
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3. Blood & Oxygen Transport

- **Hemoglobin:** Protein in red blood cells that binds oxygen.
 - **SpO₂ (Oxygen Saturation):** Percent of hemoglobin carrying oxygen. Normal: **95–100%**.
 - **Hypoxia:** Low oxygen in tissues; dangerous if SpO₂ < 90%.
 - **Relevance for Devices:**
 - **Pulse oximeters** → use red/IR light to measure SpO₂ via absorption.
 - **Blood analyzers** → measure O₂/CO₂ and pH levels.
 - **Wearables** → detect oxygen drops during sleep (apnea detection).
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4. Nervous System (Control & Signals)

- **Central Nervous System (CNS):** Brain + spinal cord, coordinates all body functions.

- **Peripheral Nervous System (PNS):** Nerves that transmit signals to/from CNS.
 - **Neurons:** Transmit signals via electrical impulses + neurotransmitters.
 - **Relevance for Devices:**
 - **EEG** → records brain waves for sleep studies, seizure detection.
 - **EMG** → measures muscle activation for prosthetics or rehab.
 - **Neurostimulators** → modulate brain/spinal cord activity for therapy.
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5. Musculoskeletal System (Movement & Force)

- **Muscles:** Contract to produce movement, controlled by motor neurons.
 - **Bones & Joints:** Provide structure, leverage, and mobility.
 - **Relevance for Devices:**
 - **Prosthetics** → rely on EMG (muscle signals) for control.
 - **Exoskeletons** → assist with movement, rehabilitation.
 - **Force sensors** → measure grip strength or gait dynamics.
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6. Vital Signs – The Core 4 Every Device Watches

1. **Heart Rate (HR)** – beats/min.
2. **Respiratory Rate (RR)** – breaths/min.
3. **Blood Pressure (BP)** – systolic/diastolic pressure.
4. **Oxygen Saturation (SpO₂)** – % of oxygen in blood.

These four are the **baseline metrics** for almost all medical devices and clinical monitoring.

7. Why This Matters for Research/Biology Role

- Understanding **body systems** = knowing what your device should measure.
- Guides **sensor selection** (PPG vs ECG vs accelerometer).
- Helps ensure **data is clinically meaningful** (e.g., 80 bpm is normal at rest but high during sleep).
- Supports **interdisciplinary communication** with engineers → biology role explains *why* features matter, engineers handle *how*.